

JC2002 Java Programming - Practical 4 (Day 5)

The goal of this practical is to get more experience with defining methods and classes in Java, as well as using Maven and JUnit for automated testing of your Java programs.

1. Create a file **'Calculator.java'** which will define the **Calculator** class. Within this class, define three public methods: **add()**, **subtract()**, and **multiply()**. Each of these methods should take two integer parameters and return the result of the corresponding arithmetic operation as an integer. Remember to add comments to your code.
2. Create file **'RunApp.java'** which will contain a class **RunApp** with **main()** method. Use this method to create an instance of the calculator. Use the Scanner object to receive an input from the console so the user can perform the calculations. The flow should be as follows:
 - User enters the first number (followed by enter).
 - User enters operator '+', '-', or '*' (followed by enter) to decide which method of the calculator object will be called. Hint: you can use **scanner.next().charAt(0)**; to read the first character of the user input from the keyboard into a *char* variable (**scanner** is an instance of **Scanner** for system input).
 - User enters the second number (followed by enter).
 - The program returns the result of the selected operation.

Note the program should run indefinitely (i.e., the user can perform multiple calculations) until 'q' or 'Q' (as "quit") is typed as operator.